
Access-Induced Semantic Divergence in Generated Program Transformations

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Abstract

1 Access-induced semantic divergence occurs when an operation that appears ob-
2 servational updates latent access state and changes a later externally visible result.
3 We formalize this pattern as observation-sensitive deterministic systems (OSDS),
4 prove bounded structural properties for a straight-line core, and evaluate it on real
5 Python packages and generated program transformations. The real-code study
6 confirms 20 divergences across 12 unmodified packages, with 9 caller-level branch
7 effects and 19/19 mechanism controls eliminating divergence. A frozen primary
8 coding-agent benchmark contains 13 base tasks and 26 normal/warned variants.
9 Across three Codex task-model configurations, five model-task outputs were ver-
10 ified OSDS failures, all concentrated in two underlying package witnesses. The
11 benchmark ordinary checks missed all five; OSDS-aware evaluation detected all
12 five; mechanism-neutralizing controls reproduced and removed 5/5. A prospec-
13 tively frozen expansion added 7 base tasks and 14 variants. Auditing the 42
14 prospective outputs separately, 23/42 were task-compliant transformations, 19/42
15 returned unchanged code, and 0 produced verified OSDS divergence. The re-
16 sults support a bounded conclusion: generated transformations can silently vio-
17 late access-sensitive semantics on validated witnesses, and OSDS-aware replay
18 plus causal controls provide a verifiable method for detecting and attributing this
19 failure mode.

20 1 Introduction

21 Program transformations often treat logging, representation, inspection, caching, and repeated ac-
22 cess as behavior-preserving. That treatment is valid when the accessed operation is observational in
23 the semantic sense. It can fail when the operation updates latent state that a later read exposes. We
24 call this pattern access-induced semantic divergence.

25 The phenomenon is related to observer effects and probe effects, but the paper studies a specific
26 sequential mechanism: an observation-shaped access mutates latent access state, and a later program
27 result changes. This matters for generated code because requested edits are often local and plausible.
28 A model may add instrumentation, memoize a representation, or simplify repeated access while
29 preserving ordinary tests that do not vary observation order.

30 We study this mechanism through three layers. The first layer is an observation-sensitive
31 deterministic-system (OSDS) model. It separates exposed read values from latent access state
32 and supports formal statements about determinism, zero-divergence conditions, and preservation
33 of body-level divergence under a linear cap. The second layer is a real-code oracle over unmodified
34 Python packages. It confirms executable package witnesses and caller-level effects. The third layer
35 is a frozen generated-transformation study built from those witnesses.

```

Original shape:
  with catching_logs(handler, level):
    result = subject(logger, handler)

Generated instrumentation shape:
  logger.debug("diagnostic")
  with catching_logs(handler, level):
    result = subject(logger, handler)

```

Figure 1: Compact pytest failure schema. The inserted log call looks like observation. In the verified Terra and Luna rows, it mutates logging/handler state that is read later. Ordinary benchmark checks passed; the OSDS oracle failed; isolating the diagnostic logger from the captured handler hierarchy removed the divergence for all four pytest rows.

36 The empirical claim is bounded. We do not estimate ecosystem prevalence, and the three
 37 tested Codex task-model configurations are not independent model providers. The main result is
 38 mechanism-grounded: in the primary benchmark, five generated outputs silently violated OSDS
 39 preservation under ordinary checks, and all five failures disappeared when the identified access-
 40 induced mechanism was neutralized while the generated code stayed fixed.

41 **Contributions.** We make four contributions. First, we give an OSDS formal model for access-state-
 42 carrying sequential systems. Second, we report 20 confirmed real-package divergences across 12
 43 packages, including 9 caller-level branch effects. Third, we evaluate frozen generated transforma-
 44 tions and identify five silent OSDS failures across Terra and Luna configurations, with zero false
 45 YES self-assessments in the primary study. Fourth, we correct the prospective expansion account-
 46 ing by separating task compliance, ordinary correctness, and OSDS preservation: 23/42 prospective
 47 outputs were task-compliant transformations, 19/42 were unchanged, and 0/42 produced verified
 48 OSDS divergence.

49 2 Motivating Real-Package Example

50 Figure 1 shows the shortest verified failure family from the primary study. The task asked for
 51 behavior-preserving instrumentation around a pytest logging witness. The generated edit inserted
 52 diagnostic logging that looked observational. In the witness, that access interacted with the captured
 53 handler hierarchy used by `catching_logs`; the later handler-level behavior changed.

54 The example gives the intuition before the formal notation. The generated transformation did not
 55 crash, did not fail the ordinary smoke check, and did not look like a destructive update at the edit
 56 site. The divergence appeared only when the evaluation compared the later visible result across the
 57 access order that the witness exposes. In the causal-control replay, the exact generated files were
 58 reused. The only change was the witness environment: diagnostic logging was isolated from the
 59 handler hierarchy responsible for the latent mutation. Under that neutralized mechanism, the OSDS
 60 divergence disappeared.

61 The state flow is compact:

```

    logger.debug → handler/logging state changes,
                  → catching_logs reads that state,
                  → caller-visible result changes.

```

62 This is a generated transformation failure, not a defect claim about pytest. It shows that a preserva-
 63 tion checker for generated edits needs to model the latent state that observation-shaped accesses can
 64 affect.

65 3 Observation-Sensitive Deterministic Systems

66 An OSDS configuration contains a semantic value $x = (b, a, d)$ and an accumulator y . The compo-
 67 nent b is stable, a is an access count, and d is latent drift. A read exposes a value and updates access
 68 state:

$$\text{read}(b, a, d, y) = ((b, a + 1, r(a, d)), y + f(b, a, d)).$$

Stage	Baseline path	Transformed path	OSDS question
Initialize	matched object	matched object	same visible setup
Observe	no extra access	observation-shaped access	can latent state drift?
Read	later visible read	later visible read	does exposed value change?
Classify	ordinary check	OSDS comparison	is divergence access-induced?

Figure 2: OSDS evaluation compares access-order variants that ordinary single-order checks may not exercise.

69 An observation exposes no additive value but can update latent drift:

$$\text{obs}(b, a, d, y) = ((b, a, g(d)), y).$$

70 A program body is a deterministic fold over a finite operation sequence $\pi \in \{\text{read}, \text{obs}\}^*$. A cap C
71 maps the final body configuration to the externally visible result.

72 For a fixed initial configuration cfg_0 , body sequence π , and cap C , let $\llbracket \pi \rrbracket_C(\text{cfg}_0)$ denote the final
73 visible output. Given a multiset of operations M and an admissible ordering set $\Pi(M)$, semantic
74 divergence is

$$\Delta_C(M, \text{cfg}_0) = \max_{\pi \in \Pi(M)} \llbracket \pi \rrbracket_C(\text{cfg}_0) - \min_{\pi \in \Pi(M)} \llbracket \pi \rrbracket_C(\text{cfg}_0).$$

75 A transformation is OSDS-preserving for a witness when the baseline and candidate agree under
76 the witness’s OSDS relation. In the benchmark, this relation is implemented as a package-specific
77 metamorphic oracle and then manually reviewed for access-induced attribution.

78 We use two evidence roles. Hidden-observation rows contain accesses that look observational in
79 the requested edit and reveal their effect only through later state. Expected-access-sensitive rows
80 contain operations whose consumption or mutation should be visible from the code shape. The
81 primary verified OSDS failures all occurred in hidden-observation rows.

82 4 Structural Results

83 The formal core is a straight-line deterministic template. Within that scope, fixed-order execution
84 is deterministic: induction over the operation sequence gives a unique next configuration at each
85 step. Two zero-divergence conditions follow directly. If $g(d) = d$, observations are inert and can be
86 moved without changing the reduced read sequence. If $f(b, a, d) = h(b)$, reads ignore access count
87 and drift, so every ordering with the same number of reads produces the same body accumulator.

88 A linear cap preserves body-level divergence. For $C(y) = \alpha y + \beta$ with $\alpha \neq 0$, $y_1 \neq y_2$ implies
89 $C(y_1) - C(y_2) = \alpha(y_1 - y_2) \neq 0$. This proposition is the only cap theorem used as an unbounded
90 formal claim. Nonlinear amplification and broad degree laws are treated as bounded computational
91 findings.

92 The appendix states a restricted adjacent-swap result for the monotone positive-parameter fragment.
93 Under deterministic updates, monotone positive observation drift, and monotone access-sensitive
94 reads, extrema over the checked order set occur at boundary orderings obtained by adjacent swaps
95 that move observations before or after reads. The appendix also states the restricted range-degree
96 result supported by exact rational checks: for checked compositional OSDS cases $m = 1..4$, cap
97 degree $d = 1..5$, and $L = 2..15$, extremal-order interpolation detects degree $2d$. This is not a
98 theorem for all OSDS programs.

99 5 Real-Code Study

100 The real-code study uses static screening only as a review queue. The screen covered 73 PyPI pack-
101 ages and 278 reviewed MEDIUM/HIGH findings. Manual review labeled 203 likely true positives
102 and 75 likely false positives, for reviewed precision 0.7302. A rebuilt exact-version source snapshot
103 contained 4,383 analyzable classes, and a deterministic sample of 200 unflagged classes found 0
104 likely missed matches. These numbers are audit evidence rather than prevalence evidence.

105 Executable confirmation is the decisive filter. The oracle selected 60 candidates, constructed 39
106 executable package-shaped harnesses, and confirmed 20 divergences across 12 unmodified pack-

Real-code evidence item	Count
Reviewed static findings	278
Executable harnesses	39
Confirmed divergences	20
Confirmed packages	12
Caller-level wrappers changing downstream execution	9
Mechanism controls eliminating divergence	19/19

Table 1: Real-code OSDS evidence. Static screening is a review queue; executable harnesses and controls provide the semantic evidence.

Model	Exec.	Pres.	Ord. bugs	Invalid	Ver. OSDS	Silent
gpt-5.6-sol	26	23	3	0	0	0
gpt-5.6-terra	24	18	4	2	2	2
gpt-5.6-luna	24	17	4	2	3	3

Table 2: Primary frozen coding-agent results. All verified OSDS failures passed the benchmark ordinary checks and failed OSDS-aware evaluation.

ages: httpcore, PyYAML, pytest, markdown, more-itertools, docutils, beautifulsoup4, boltons, cerberus, dnspython, h11, and anyio. The divergences include stream materialization, identity-cache reuse, handler-level mutation, reference-registry mutation, cursor advance, destructive tree extraction, cache recency/statistics, validation-error population, and buffer consumption.

Nine confirmed divergences were lifted into caller-level wrappers that changed branch labels and downstream consequences. Nineteen controls eliminated divergence under fresh-object, reset-between, or pure-observation interventions. These controls show that the detected differences depend on the access-induced mechanism named by the witness.

6 Generated Transformation Study

The primary benchmark was frozen before model execution. It contains 13 base tasks and 26 normal/warned variants from 9 packages and 9 unique witnesses. The six transformation families are instrumentation, caching/materialization, refactoring, repeated-access cleanup, access reordering, and debugging/inspection. Each generation task was fresh and projectless. The model saw only the frozen prompt, with no OSDS labels, oracle output, witness labels, benchmark explanation, or previous responses.

Replay treats generated text as data. Raw responses are saved exactly, code is extracted, ordinary smoke checks are run, and then OSDS-aware checks compare baseline and candidate behavior under the witness relation. Manual review classifies non-preserved rows as verified semantic divergence, ordinary programming bug, invalid patch, environment failure, oracle issue, or unclear.

The five verified OSDS failures are five model-task outcomes over two underlying package witnesses. Terra failed on two pytest instrumentation rows. Luna failed on those two pytest rows and one PyYAML caching/materialization row. Sol preserved the verified-OSDS rows on which Terra or Luna diverged. Expected-access-sensitive rows produced ordinary bugs or invalid patches, with no verified OSDS divergence.

Self-assessment was secondary. Across Sol, Terra, and Luna, there were zero false YES preservation claims on verified OSDS divergences. Conservative NO responses were common, so self-assessment should not replace execution replay.

7 Causal Attribution and Prospective Expansion

The causal-control experiment replays the exact generated files for the five verified failures. The candidate source remains unchanged. The intervention changes only the witness mechanism that caused access-induced divergence.

Family	Intervention	Rows	Result
pytest instrumentation	isolate diagnostic logger from captured handler hierarchy	4	4/4 removed
PyYAML caching	clear representer identity cache after observation	1	1/1 removed

Table 3: Causal controls. Under the original witness all five failures reproduced; under mechanism-neutralizing controls all five OSDS divergences disappeared.

Model	Outputs	Task-compliant	Unchanged	Ordinary pass	Ver. OSDS
gpt-5.6-sol	14	8	6	14	0
gpt-5.6-terra	14	8	6	14	0
gpt-5.6-luna	14	7	7	14	0
Total	42	23	19	42	0

Table 4: Prospective expansion with task compliance separated from semantic preservation. Unchanged outputs are OSDS-preserving but are not counted as task-compliant transformations.

138 This supports the access-induced attribution. The failures are mechanism-dependent generated-code
139 defects: ordinary checks pass, OSDS checks fail, and targeted neutralization removes divergence
140 while preserving the generated candidate.

141 After the causal phase, 11 unused confirmed real-code witnesses were audited prospectively. Seven
142 met eligibility criteria and were frozen before model execution, producing 14 normal/warned vari-
143 ants from boltons, dnspython, and h11. Sol, Terra, and Luna were run on these exact prompts as
144 fresh projectless tasks. Self-assessment was skipped for the expansion.

145 The prospective benchmark produced no new verified OSDS divergences. The corrected interpre-
146 tation is narrower than a 42/42 transformation success claim: 23/42 outputs performed a nontrivial
147 task-compliant transformation, 19/42 returned unchanged code, and 23/23 task-compliant outputs
148 were OSDS-preserving. The result constrains failure incidence among these frozen responses and
149 supports reporting the primary failures as reproducible examples rather than as a prevalence esti-
150 mate.

151 8 Related Work

152 **Observer effects and Heisenbugs.** Heisenbugs and probe effects describe programs whose behav-
153 ior changes under observation or debugging. Musuvathi et al. discuss debugging statements affect-
154 ing concurrent interleavings in systematic testing [Musuvathi et al., 2008]. Sallinger, Weissenbacher,
155 and Zuleger formalize Heisenbugs as hyperproperties and analyze causes including probe effects and
156 nondeterminism [Sallinger et al., 2026]. OSDS is narrower: it models sequential access-state mecha-
157 nisms where observation-shaped operations mutate latent state and later reads expose the difference.

158 **Semantic preservation and equivalence.** Compiler verification and translation validation study
159 semantic preservation between source and transformed programs [Pnueli et al., 1998, Leroy, 2009].
160 OSDS uses the same preservation motivation but focuses on generated source-level transformations
161 over library objects whose read and observation operations carry latent state. The oracle compares
162 access-order behavior rather than proving full program equivalence.

163 **Metamorphic testing.** Metamorphic testing addresses oracle problems by checking relations across
164 related executions [Chen et al., 1998, 2018]. The OSDS-aware oracle is a specialized metamorphic
165 relation: the follow-up execution changes the placement of an observation-shaped access while
166 keeping witness setup matched. Generic metamorphic testing supplies the testing pattern; OSDS
167 supplies the mechanism and classification boundary.

168 **Current generated-transformation benchmarks.** Differential fuzzing has recently been applied
169 to LLM-generated code refactorings, showing that generated refactors can be functionally nonequiv-
170 alent and that existing tests can miss a substantial fraction of those cases [Dristi and Dwyer,
171 2026]. SWE-Refactor moves refactoring evaluation to repository-level Java changes mined from

real projects, emphasizing realistic context and compound refactorings [Xu et al., 2026]. VERINA benchmarks verifiable code generation in Lean across code, specification, and proof generation tasks [Ye et al., 2025]. Viverra studies text-to-code with generated assertions and bounded verification, surfacing verified properties alongside generated C programs [Wu et al., 2026]. Metamorphic transformations have also been used to diagnose memorization in LLM-based program repair; we use that line only as related evidence that semantics-preserving transformations can support stronger code-evaluation protocols [Koning et al., 2026].

OSDS mechanism attribution. Generic differential and metamorphic testing can establish that two executions differ under a chosen relation. OSDS contributes a sequential semantic model for objects whose reads and observations interact through latent access state. That structure determines which access reorderings form meaningful witnesses and supports mechanism-specific controls. The causal interventions test whether a detected divergence disappears when the access-induced mechanism is neutralized while the generated candidate remains unchanged, giving mechanism attribution alongside behavioral detection.

Generated code and repair verification. HumanEval and Codex made execution-based evaluation central for code generation [Chen et al., 2021]. SWE-bench extends evaluation to issue-resolution patches in real repositories [Jimenez et al., 2024]. Automated program repair has long shown that patches can overfit tests [Le Goues et al., 2012, Smith et al., 2015]. This paper studies a different preservation failure: a generated transformation can pass the ordinary benchmark tests while violating an access-state relation that those tests do not exercise.

9 Limitations

The coding-agent benchmark is small and correlated by witness and package. Counts are benchmark outcomes, not prevalence estimates. The five verified primary failures are five model-task failures over two underlying package witnesses. The prospective expansion added seven witnesses and found zero new verified OSDS divergences.

The three model labels are Codex task-model configurations reported by the Codex runtime. They were collected through fresh projectless Codex tasks rather than the OpenAI API, and temperature/seed were not exposed. The study is Python-focused. Manual review remains part of verified OSDS classification. The formal results apply to the stated straight-line deterministic core; broader analyzer and range-degree claims remain empirical or bounded computational evidence.

The intended positive impact is better preservation testing for generated code in libraries where observation-shaped accesses can mutate hidden state. The main risk is that the same examples could encourage superficial transformations that pass ordinary tests while changing access-sensitive behavior. We mitigate this risk by releasing bounded witnesses, replay scripts, and mechanism-neutralizing controls rather than a general attack system, and by framing the results as verification evidence rather than deployment guidance.

10 Conclusion

Access-induced semantic divergence is a concrete preservation risk for generated transformations on access-state-carrying software objects. The OSDS model identifies the mechanism, the real-code study confirms package witnesses, and the coding-agent benchmark shows five silent generated failures under ordinary checks. Exact replay plus mechanism-neutralizing controls reproduced and removed all five primary failures. The prospective expansion found no new OSDS divergences and exposed substantial unchanged-output behavior. The resulting assessment is bounded: OSDS-aware verification provides a reproducible method for detecting and attributing a semantic failure mode that ordinary checks missed in this benchmark, without supporting broad claims of model unsafety.

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259 A Formal Details

260 A.1 State and transitions

261 An OSDS state is $s = (b, a, d, y)$ with base b , access count a , latent drift d , and accumulator y . A
 262 read transition is deterministic:

$$R(s) = (b, a + 1, r(a, d), y + f(b, a, d)).$$

263 An observation transition is deterministic:

$$O(s) = (b, a, g(d), y).$$

264 For a finite sequence $\pi = o_1 \dots o_n$, composition is $\llbracket \epsilon \rrbracket(s) = s$ and $\llbracket \pi o \rrbracket(s) = o(\llbracket \pi \rrbracket(s))$. Branch
 265 evaluation is a deterministic predicate $B(C(\llbracket \pi \rrbracket(s)))$ over the capped visible result or final configu-
 266 ration. Branch divergence occurs when two admissible orderings produce different branch labels or
 267 downstream consequences.

268 A.2 Semantic divergence

269 Given operation multiset M , admissible orderings $\Pi(M)$, initial state s_0 , and deterministic cap C ,
 270 define $V_C(\pi, s_0) = C(\llbracket \pi \rrbracket(s_0))$. The access-order range is $\Delta_C(M, s_0) = \max_{\pi \in \Pi(M)} V_C(\pi, s_0) -$
 271 $\min_{\pi \in \Pi(M)} V_C(\pi, s_0)$ for finite $\Pi(M)$. A transformation preserves the witness when baseline and
 272 candidate have equal OSDS-observable results under the witness relation.

273 A.3 Fixed-order determinism

274 **Proposition.** For fixed s_0 , fixed π , deterministic R , deterministic O , and deterministic C , $V_C(\pi, s_0)$
 275 is unique. **Proof.** Induct on $|\pi|$. The empty sequence returns s_0 . For πo , the induction hypothesis
 276 gives a unique intermediate state and $o \in \{R, O\}$ maps it to a unique next state. The deterministic
 277 cap gives a unique output.

278 A.4 Zero divergence under no observation-induced mutation

279 **Proposition.** If $g(d) = d$ and observations expose no additive value, moving observations within a
 280 fixed multiset does not change the body accumulator or final latent state. **Proof.** Each observation
 281 is the identity on (b, a, d, y) and contributes no value to the accumulator. Delete observations from
 282 any admissible ordering; the remaining read sequence has the same length and order because all
 283 reads are identical. The same deterministic reads are therefore executed from the same state and
 284 contribute the same accumulator terms. Re-inserting identity observations at arbitrary positions
 285 leaves the accumulator and final latent state unchanged.

286 A.5 Zero divergence under access-insensitive reads

287 **Proposition.** If $f(b, a, d) = h(b)$, every ordering with L reads has the same body accumulator
 288 $Lh(b)$. **Proof.** Every read contributes the same exposed value for fixed b regardless of access count
 289 or drift. Observations may change d , but d is ignored by f . A deterministic cap depending only on
 290 the accumulator and final counts preserves equality.

291 A.6 Linear-cap preservation

292 **Proposition.** Let $C(y) = \alpha y + \beta$ with $\alpha \neq 0$. If body accumulators $y_1 \neq y_2$, then $C(y_1) \neq C(y_2)$.
 293 **Proof.** $C(y_1) - C(y_2) = \alpha(y_1 - y_2)$, which is nonzero under exact arithmetic.

294 A.7 Adjacent-swap extrema under monotone access drift

295 **Restricted lemma.** Consider the positive monotone fragment with identical reads, identical ob-
 296 servations, deterministic updates, observation update g satisfying $g(d) \geq d$, and read contribution
 297 $f(b, a, d)$ monotone nondecreasing in d for fixed b, a . Assume observations do not change a or y ,
 298 reads increase a by one, and the final cap is monotone in the accumulated read contribution. For any
 299 adjacent pair RO in an otherwise fixed ordering, replacing it by OR cannot decrease the contribu-
 300 tion of the swapped read or any later read. **Proof.** Let the common prefix before the pair produce
 301 state (b, a, d, y) . In the local order RO , the read contributes $f(b, a, d)$ and the following observation
 302 changes drift to $g(d)$. In the local order OR , the observation first changes drift to $g(d)$, and the read
 303 contributes $f(b, a, g(d))$. Since $g(d) \geq d$ and f is monotone in drift, $f(b, a, g(d)) \geq f(b, a, d)$;
 304 the read count after the pair is the same in both orders. The suffix after the pair therefore starts
 305 with the same b and a , an accumulator no smaller in the OR run, and drift $g(d)$ in both runs. In-
 306 ducting over later operations gives no smaller drift-visible read contributions and no smaller final
 307 capped value. Repeated adjacent swaps can move each observation left across neighboring reads
 308 without decreasing value, so an ordering with all observations before all reads is a maximum among
 309 orderings reachable by these swaps. The reverse argument moves observations right across reads
 310 without increasing value, giving the all-reads-before-observations boundary as a minimum. **Scope.**
 311 The lemma is limited to this positive monotone straight-line fragment with identical operations and
 312 monotone caps. It is not claimed for arbitrary Python programs or nonmonotone observation effects.

A.8 Restricted 2d range-degree statement

Bounded exact computational result. For the compositional OSDS cap family implemented in the exact-rational validation scripts, with $m = 1.4$, cap degree $d = 1.5$, and $L = 2.15$, exhaustive checks on small feasible grids plus extremal-order interpolation detect access-order range degree $2d$. **Derivation.** In the checked family, the body-range term is $\eta mL(L - 1)/2$. The compositional cap contributes $d - 1$ final read factors, each quadratic in L . The leading degree is therefore $2 + 2(d - 1) = 2d$. Exact rational scripts verify the extremal formula against enumerated feasible orderings and holdout values on the stated bounded domain. **Scope.** This is a restricted exact computational result for the named family and grid. It is not a general theorem for all OSDS programs.

B Benchmark Details

Primary tasks are in `benchmark/tasks.jsonl` under the agent-behavior study tree. Primary prompt files are in that tree's `prompts/` directory. The pre-model manifest and raw Codex task-model responses are in `external_collection/` and `external_collection/responses/`.

Manual review rows are in the primary study analysis directory as the 20260813 manual-review JSONL. The matching cross-model summary is stored beside it as JSON and Markdown. Prospective tasks, prompts, and raw responses are under `benchmark_expansion/`. Exact prospective replay outputs are in the three `*expansion-exact-20260813Tcutscope` result directories.

The prospective task-compliance audit is `analysis/prospective_task_compliance.csv`. It separates `task_compliance`, `ordinary_correctness`, and `osds_preservation` for all 42 prospective outputs. The supplement artifact index maps these shortened descriptions to exact relative paths.

C Reproducibility

The reviewer supplement contains a relative-path artifact index, frozen prompts, raw response JSONLs, replay summaries, causal-control scripts, real-code oracle metadata, and expected outputs. It excludes virtual environments, caches, authentication material, and personal absolute paths. Replays require Python 3.11 or newer with `pytest` and the package versions named by the witness manifests. The supplement `REPRODUCE.md` gives exact commands for the primary replay, prospective replay, causal-control replay, and table reproduction.

All artifact replays are CPU-only Python executions and do not require a GPU, accelerator runtime, cloud worker, model credential, or model-provider API call. A pre-submission reproduction check on Windows 11 with Python 3.14.4, a 6-core/12-thread AMD Ryzen 5 4600H CPU, and 7.4 GiB RAM measured 1.12 s for a baseline single-task replay, 1.22 s for a frozen primary single-task replay, 1.13 s for a frozen prospective single-task replay, 3.49 s for the five-case causal-control replay, 0.57 s for the real-code witness validation command, and 0.13 s for the summary-table command. Exact peak memory was not logged; the commands operate on small JSONL, CSV, and Python files, and the packaged supplement is below 1 MB. The earlier Codex task-model generation phase is not reproduced by the supplement. Its inference hardware, accelerator hours, and hidden service-side memory were not exposed by the Codex task interface, and preliminary or failed collection attempts used additional model calls outside the replay package.

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